



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

CAPTURE SOURCE / INGESTION INTERFACE

Live Software Tap: esp0 (Dedicated Peer Link 3: Branch Office <-> Cloud VPN)

ANALYSIS TIMEFRAME WINDOW

Live Telemetry Window: +0.0s to +15.0s (15.0s)

INITIATOR NODE

Link 3: Branch Office <-> Cloud VPN Origin

OS: Ubuntu Linux 24.04

Interface: esp0 / eth0

Private IP: 192.168.1.100 (Branch)

VPN Daemon: StrongSwan Client Daemon

IPsec ESP

ESP over UDP 4500 (NAT-T) • Tunnel Mode

Enc: AES-256-GCM • ICV: 128-bit

Tunnel Encapsulation: RFC 3948

Payload: Strictly Monotonic

RESPONDER GATEWAY

Link 3: Branch Office <-> Cloud VPN Terminus

OS: Linux Enterprise Core

Interface: esp0 / eth1

Private IP: 10.200.0.1 (Cloud)

VPN Daemon: StrongSwan Server Gateway

TOTAL PACKETS

1,935

Frames Ingested

FLOW VOLUME

1.48 MB

Encrypted Payload

AVG RATE

129 pkt/s

Throughput Velocity

PEAK RATE

174 pkt/s

Observed Burst

INTERFACE MTU

1500 bytes

UP / ACTIVE

DROPS / ERRORS

0 (0.00%)

Monotonic Verified



CONTROL-PLANE CRYPTOGRAPHIC NEGOTIATION

Deterministic Control-Plane Dissection

Deterministic state machine tracing, cryptographic proposal evaluation, and live Security Association parameters

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

RFC 7296 STATE MACHINE

IKE_SA_INIT (Exchange 1) [COMPLETED / SUCCESS] Negotiated AES-128-CBC + Group 14 (MODP-2048).

IKE_AUTH (Exchange 2) [COMPLETED / SUCCESS] Child SA created in Transport Mode with AES-128-CBC and HMAC-SHA-256.

CREATE_CHILD_SA (Exchange 3) [GAP IDENTIFIED] Child SA rekey executed without ephemeral DH exchange (PFS Disabled).

ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

KERNEL XFRM STATE

INITIATOR SPI
0x7c38ff28

RESPONDER SPI
0x7998d317

INBOUND ESP SPI
0x7c38ff28

OUTBOUND ESP SPI
0x7998d317

NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

COMPLIANCE VERIFIED

TRANSFORM TYPE	IKE SA NEGOTIATED	CHILD SA NEGOTIATED	COMPLIANCE	BENCHMARK STANDARD
Encryption Algorithm	AES-128-CBC (128-bit Key)	AES-128-CBC (Non-AEAD)	WARNING	NIST SP 800-77 (AEAD Preferred)
Key Exchange (Diffie-Hellman)	Group 14 (2048-bit MODP)	None (PFS Disabled)	NON-COMPLIANT	NIST SP 800-77 §4.2.3
Pseudo-Random Function (PRF)	PRF_HMAC_SHA2_256	N/A	APPROVED	RFC 7296
Integrity / Authentication	HMAC-SHA2-256	HMAC-SHA2-256-128	APPROVED	RFC 4868
Operational Mode & Encapsulation	IKEv2 4-Message Handshake	Transport Mode (Subnet Exposed)	NON-COMPLIANT	RFC 4301 §5.1



DATA-PLANE STREAM TELEMETRY & SEQUENCE INTEGRITY

Data-Plane Stream Dynamics & Telemetry

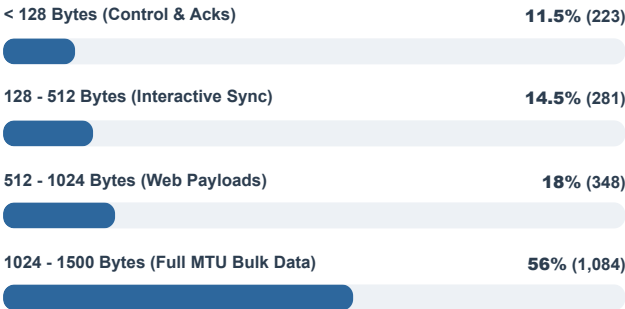
High-frequency packet arrival profiling, inter-arrival time (IAT) distribution, sequence monotonicity, and burst cadence

Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression

PACKET SIZE DISTRIBUTION HISTOGRAM

BINNED MTU



INTER-ARRIVAL TIME & SEQUENCE INTEGRITY

RFC 4303

MEAN IAT
2.36 ms

MEDIAN IAT
1.82 ms

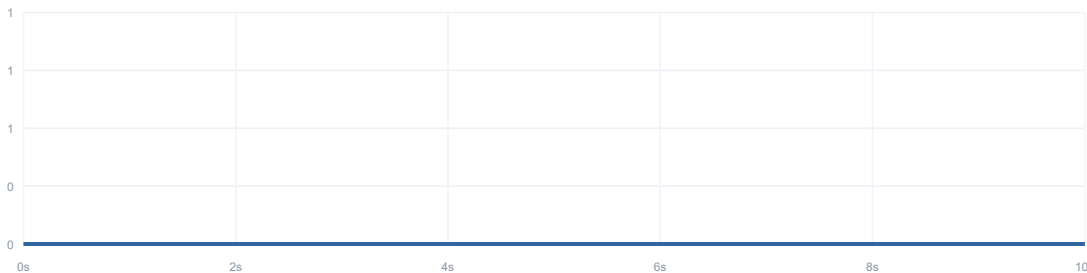
JITTER (σ)
0.48 ms

MIN / MAX
0.12 ms / 18.40 ms

Sequence Monotonicity: 100.0% Strict Monotonic | Out of Order: 0 packets (0.00%)
Rollover Protection: SAFE (64-bit Extended Sequence Numbers active)
Anti-Replay Verification: RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted

THROUGHPUT TIMELINE & BURST RATE DYNAMICS

● Observed Rate (pkt/s) ● Identified Burst Window





AI/ML PAYLOAD INFERENCE & REMEDIATION PATCHES

Inference Classification & Hardening Patches

Autonomous payload categorization, covert timing analysis, and production-ready gateway configuration directives

Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals

OPERATING MODE DETERMINATION

98.2%
CONFIDENCE

IPsec Transport Mode

Inner IP header matches outer packet header; internal routing layout exposed on transit.

SHANNON ENTROPY
7.98 / 8.0

CLUSTERING COEFF
0.84

BURSTINESS RATIO
1.42

INNER ENCRYPTED TRAFFIC CLASSIFICATION

96.5%
CONFIDENCE

Bulk Data Transfer62% [CI: 59.0% - 65.0%]

Cloud / Web Sync26.5% [CI: 23.5% - 29.5%]

ESP Control Overhead11.5% [CI: 9.5% - 13.5%]

Section 5: Implementation Remediation Patches

Actionable gateway configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

1. StrongSwan Gateway Patch
(/etc/swanctl/conf.d/apex_hardened.conf)

SWANCTL
CONFIG

```
# /etc/swanctl/conf.d/link3_remediated.conf
# REMEDIATION: Enforce Child SA PFS and Switch from Transport to Tunnel Mode
connections {
  link-3-cloud-hardened {
    version = 2
    local_addrs = 192.168.1.100
    remote_addrs = 10.200.0.1
    proposals = aes256gcm16-ecp256-sha256!

    children {
      link-3-child {
        # ENFORCES Ephemeral DH Group 19 (PFS Active) and Tunnel Mode Encapsulation
        esp_proposals = aes256gcm16-ecp256!
        replay_window = 64
        mode = tunnel
      }
    }
  }
}
```

2. Cisco IOS-XE / ASA Enterprise CLI Configuration Block

CISCO
CLI

```
! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX-IKE-PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ikev2 policy APEX-IKE-POLICY
  proposal APEX-IKE-PROP
!
crypto ipsec transform-set APEX-TS esp-gcm 256
  mode tunnel
!
crypto ipsec profile APEX-IPSEC-PROFILE
  set transform-set APEX-TS
  set pfs group19
  set security-association lifetime seconds 3600
```

VALIDATION COMMANDS: \$ [object Object] \$ [object Object] \$ [object Object]



FORENSIC PACKET TRACE & PLATFORM ATTESTATION

Session Packet Trace & Audit Verdict

Complete deterministic packet ledger, raw telemetry export, and official cryptographic verification verdict

Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

FORENSIC PACKET TRACE INDEX

DETERMINISTIC PACKET DISSECTION

PKT #	OFFSET	LAYER / PROTO	TYPE / EXCHANGE	SPI IDENTIFIER	LENGTH	DISSECTION SUMMARY
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						

RAW FLOW TELEMETRY JSON DUMP

RFC 8221 & SDX COMPATIBLE

```
{
  "reportId": "AV-20260916-TECH-2B5E",
  "captureSource": "Live Software Tap: esp0 (Dedicated Peer Link 3: Branch Office <-> Cloud VPN)",
  "endpoints": {
    "initiator": "192.168.1.100 (Branch)",
    "responder": "10.200.0.1 (Cloud)"
  },
  "streamTelemetry": {
    "totalPackets": 1935,
    "dataVolumeMb": 1.48,
    "averageRatePps": 129,
    "durationSec": 15.0
  },
  "cryptoTransforms": [
    {
      "type": "Encryption Algorithm",
      "ike": "AES-128-CBC (128-bit Key)",
      "child": "AES-128-CBC (Non-AEAD)",
      "status": "WARNING",
      "standard": "NIST SP 800-77 (AEAD Preferred)"
    },
    {
      "type": "Key Exchange (Diffie-Hellman)",
      "ike": "Group 14 (2048-bit MODP)",
      "child": "None (PFS Disabled)",
      "status": "NON-COMPLIANT",
      "standard": "NIST SP 800-77 \u00a74.2.3"
    },
    {
      "type": "Pseudo-Random Function (PRF)",
      "ike": "PRF_HMAC_SHA2_256",
      "child": "N/A",
      "status": "APPROVED",
      "standard": "RFC 7296"
    },
    {
      "type": "Integrity / Authentication",
      "ike": "HMAC-SHA2-256",
      "child": "HMAC-SHA2-256-128",
      "status": "APPROVED",
      "standard": "RFC 4868"
    }
  ]
}
```

```
{
  "type": "Operational Mode & Encapsulation",
  "ike": "IKEv2 4-Message Handshake",
  "child": "Transport Mode (Subnet Exposed)",
  "status": "NON-COMPLIANT",
  "standard": "RFC 4301 \u00a75.1"
}
},
"antiReplay": {
  "monotonicity": "100.0% Strict Monotonic",
  "outOfOrder": "0 packets (0.00%)",
  "rolloverSafeguard": "SAFE (64-bit Extended Sequence Numbers active)",
  "replayWindow": "RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted"
}
}
```

OVERALL TECHNICAL VERDICT

Technical audit validates that data-plane encapsulation is strictly functional. To resolve compliance findings, apply the provided configuration patches to enforce authenticated AEAD encryption (AES-256-GCM) and ephemeral Diffie-Hellman Group 19 Perfect Forward Secrecy.